

6. Satellites in orbit around the Earth are used for a variety of purposes. One use is communications between base stations and satellites. Satellites for communications are usually put into geostationary orbits, orbiting above the equator. Satellites for other uses such as weather forecasting can be put in polar orbits. The speed of the satellite and therefore its orbital period, the time for one orbit of the Earth, is determined by its orbit height.

- (a) Another kind of orbit is a geosynchronous orbit. State **one** similarity and **one** difference between a geostationary and a geosynchronous orbit. [2]

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- (b) The table below gives data on the orbits of different satellites.

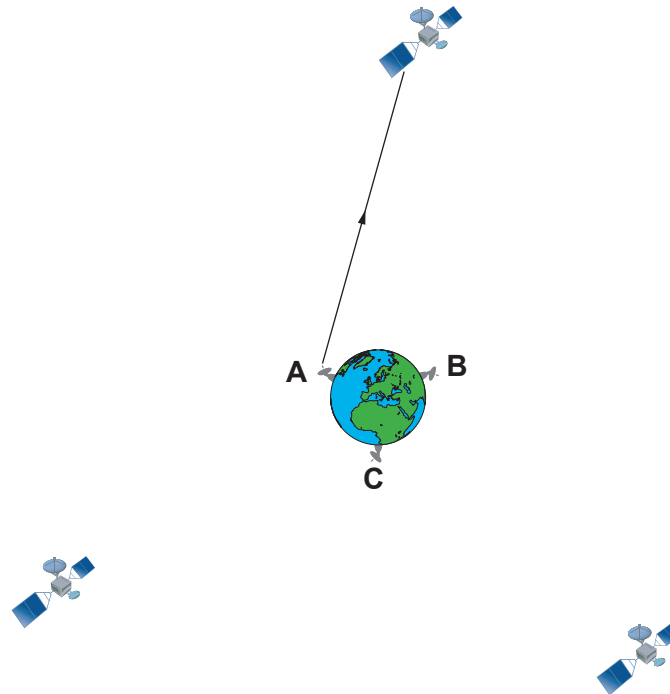
Satellite	Height above Earth (km)	Period (hours)
1	21 450	12.9
2	24 000	15
3	36 000	24
4	507	1.6
5	1 090	1.8
6	20 200	12

- (i) State which satellite is in geostationary orbit. [1]

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- (ii) A signal is sent from base station **A** to a geostationary satellite and is received at another base station after a time delay of 0.48 s. An incomplete diagram of the signal's path is shown.



Use an equation from page 2 to calculate the distance the signal has travelled. The speed of light, $c = 3 \times 10^8$ m/s. [3]

Distance = m

- (iii) Use your answer to (ii) and the information in the table to determine which base station, **A**, **B** or **C**, finally receives the signal. Show all your workings. [2]

Receiving base station =

END OF PAPER

